Model Card: Student Score Predictor

Week 10 Lab, Part 6 | INSY 8413 Introduction to Big Data Analytics, AUCA
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1. Purpose

This model predicts a student's final score (0 to 100) in the course. It is meant for lecturers and the faculty, to spot students who may need extra support early, before the final exam.

2. Data and features

The model was trained on *week10_students.csv*: 150 AUCA students, with clean data and no missing values.

- **Inputs (features):** quiz_avg, assignments_avg, attendance_pct
- **Target:** score
- **Method:** linear regression, trained on 112 students and tested on 38 unseen students (test_size=0.25, random_state=42)

3. Performance (on unseen students)

Measure	Value	What it means in plain language
MAE	3.33 points	On average, the predicted score is off by about 3 points out of 100.
R^2	0.89	Quiz, assignments and attendance explain about 89% of why students' scores differ. The other 11% comes from things the model cannot see.

In simple terms: for a student the model has never seen, its prediction is usually close to the real score, within a few points.

4. Risks

Who could be harmed by a wrong prediction?

- A capable student wrongly predicted to score low could be labelled as "at risk", which may cause stigma and lower expectations from lecturers.
- A student who really needs help could be predicted to score well and be missed, so they get no support.
- The model only sees three numbers. It does not know about illness, family problems, language, or how hard a student is working. It was trained on 150 students from one course, so it may not work well for other groups or courses.

How the output should be used

- Use it to start a conversation with a student and to offer tutoring or other support early.
- Use it together with the lecturer's own knowledge of the student.

How it should NOT be used

- Do not use it to decide grades, pass or fail, scholarships, discipline or admission.
- Do not treat a prediction as a fact about a student, or share it with others as a label.